

## Day 7: Score, SDEs & Flow Matching

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- **Score, SDEs & Flow Matching** — The continuous-time view of diffusion
- 3 hours lecture + practical
- Slides, notes, and code on the course site

- The Score Function
- Learning the Score
- Sampling with the Score
- The Continuous-Time View
- Flow Matching
- One Model, Many Views

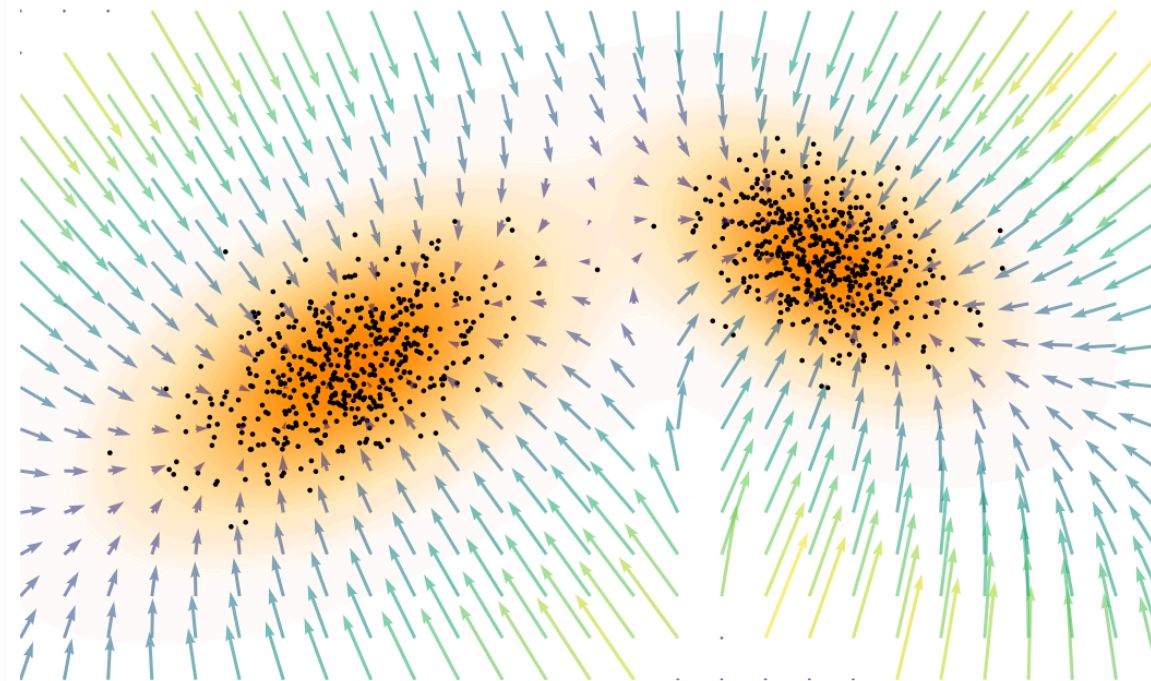
## 1 · The Score Function



## 1.1 What Is the Score?

- Score = gradient of log-density:  $s(x) = \nabla_x \log p(x)$
- A **vector field** pointing toward higher probability
- Day 6 learned to denoise; today we learn this field
- Knowing the score is enough to **sample** (Langevin, SDE, ODE)
- Source: Principles of Diffusion Models, Ch. 3–6

## 1.1 What Is the Score?



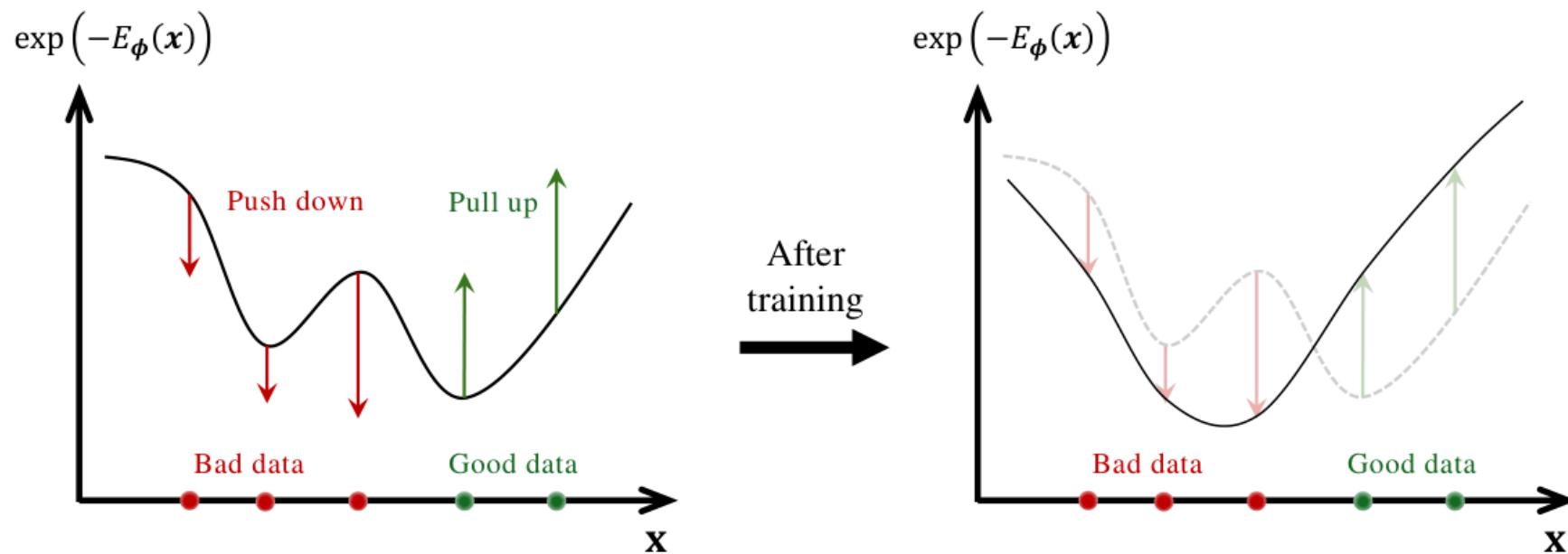
**Figure 3.2: Illustration of score vector fields.** Score vector fields  $\nabla_{\mathbf{x}} \log p(\mathbf{x})$  indicate directions of increasing density.

Source: Created by the author.

## 1.2 Energy-Based Models

- Write  $p_{\theta}(x) = e^{-E_{\theta}(x)} \frac{1}{Z_{\theta}}$
- Normalizer  $Z_{\theta} = \int e^{-E_{\theta}(x)} dx$  is intractable
- Max-likelihood needs  $\nabla_{\theta} \log Z_{\theta}$  — hard
- Idea: model the **score** and sidestep  $Z$  entirely
- Flexible energy, but sampling needs MCMC

## 1.2 Energy-Based Models



**Figure 3.1: Illustration of EBM training.** The model lowers density (raises energy) at “bad” data points (red arrows), and raises density (lowers energy) at “good” data points (green arrows).



## 1.3 Why the Score Avoids the Normalizer

- $\log p_\theta(x) = -E_\theta(x) - \log Z_\theta$
- $Z_\theta$  does not depend on  $x$
- $\rightarrow \nabla_x \log p_\theta(x) = -\nabla_x E_\theta(x)$
- The intractable constant **vanishes** under  $\nabla_x$
- Learn the shape of  $p$ , not its normalization

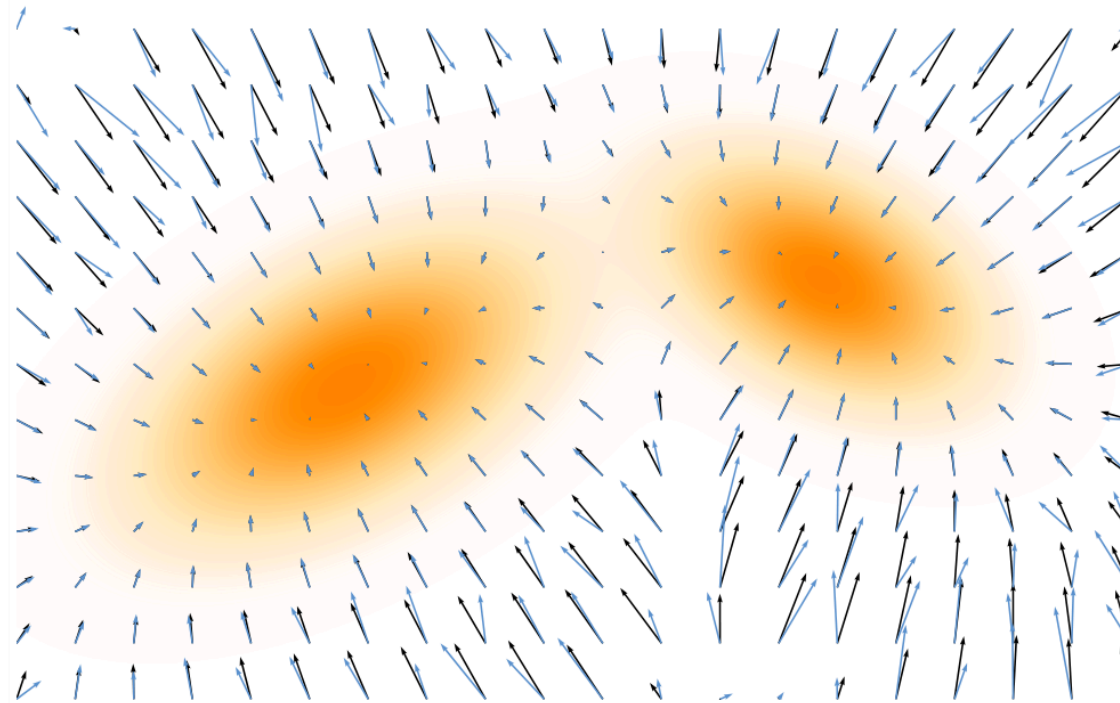
## 2 · Learning the Score



## 2.1 Score Matching

- Fit  $s_\theta(x) \approx \nabla_x \log p_{\text{data}}(x)$
- Naive loss needs the unknown true score
- Hyvärinen: integrate by parts to remove it
- $J = \mathbb{E} \left[ \frac{1}{2} \|s_\theta\|^2 + \text{tr}(\nabla_x s_\theta) \right]$
- Trace (Jacobian) term is costly in high dimension

## 2.1 Score Matching



**Figure 3.4: Illustration of Score Matching.** The neural network score  $\mathbf{s}_\phi(\mathbf{x})$  is trained to match the ground truth score  $\mathbf{s}(\mathbf{x})$  using a MSE loss. Both are represented as vector fields.

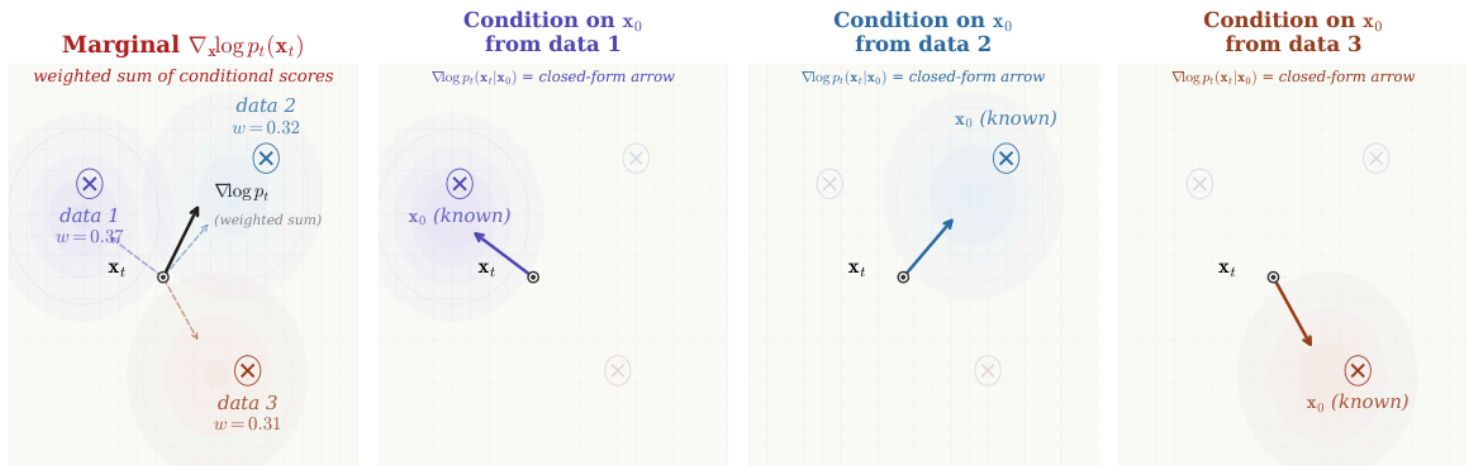
## 2.2 Derivation: Implicit Score Matching

- Start:  $\frac{1}{2}\mathbb{E}_p \left[ \|s_\theta(x) - \nabla \log p(x)\|^2 \right]$
- Expand the square; the cross term has  $\nabla \log p$
- $\mathbb{E}_p \left[ s_\theta^T \nabla \log p \right] = \mathbb{E}_{p[-\nabla \cdot s_\theta]}$  (parts)
- Uses  $p \nabla \log p = \nabla p$  and vanishing boundary
- Leaves a loss in  $s_\theta$  alone (+ const)

## 2.3 Denoising Score Matching

- Add noise:  $x_t = x_0 + \sigma\varepsilon$ , learn the **noisy** score
- Key identity:  $\nabla_{x_t} \log p(x_t \mid x_0) = -(x_t - x_0) \frac{1}{\sigma^2}$
- Target is known in closed form  $\rightarrow$  no Jacobian trace
- $J_{\text{DSM}} = \mathbb{E} \left[ \|s_\theta(x_t) + (x_t - x_0) \frac{1}{\sigma^2}\|^2 \right]$
- Score matching = denoising – the Day 6 connection

## 2.3 Denoising Score Matching



HARD: intractable weighted sum

EASY: one closed-form arrow per data point (denoising score matching)

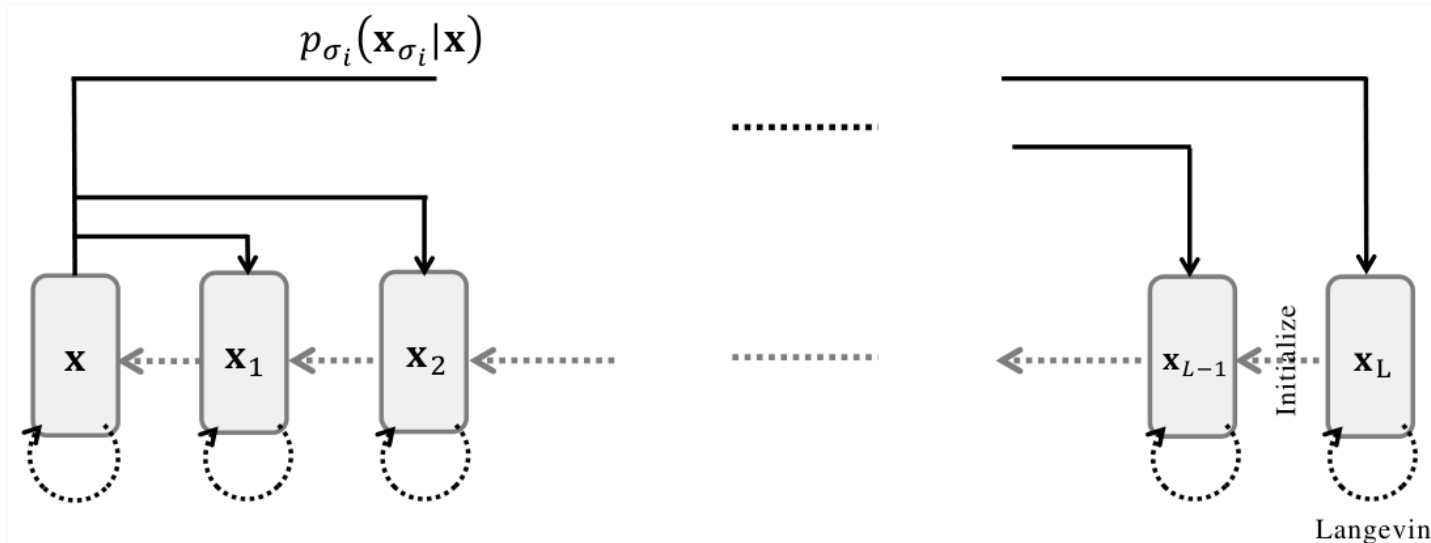
**Figure 4.6: Illustration of the denoising score matching trick.** The marginal score  $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$  is generally an intractable weighted average of the conditional scores over all possible clean samples  $\mathbf{x}_0$ , where the weights are given by the posterior distribution of  $\mathbf{x}_0$  conditioned on the noisy observation  $\mathbf{x}_t$ . However, for each fixed  $\mathbf{x}_0$ , the conditional score  $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t|\mathbf{x}_0)$  is available in closed form. Denoising score matching turns this hard marginal estimation problem into an easy per-sample regression problem.

## 2.4 Multiple Noise Scales (NCSN)

- One noise level can't cover all of space
- Low noise: accurate near data, empty regions unseen
- High noise: fills space but blurs detail
- Train **one** network conditioned on noise level  $\sigma$
- Anneal from high to low noise when sampling



## 2.4 Multiple Noise Scales (NCSN)



**Figure 3.7: Illustration of NCSN.** The forward process perturbs the data with multiple levels of additive Gaussian noise  $p_{\sigma}(\mathbf{x}_{\sigma} | \mathbf{x})$ . Generation proceeds (in dash lines) via Langevin sampling at each noise level, using the result from the current level to initialize sampling at the next lower variance.

### 3 · Sampling with the Score

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## 3.1 Langevin Dynamics

- Walk uphill in log-density, plus noise:
- $x_{k+1} = x_k + \tau s_\theta(x_k) + \sqrt{2\tau} z_k$
- Stationary distribution is exactly  $p(x)$
- Drift toward data + noise to explore
- Anneal  $\sigma$  (NCSN) for fast, stable mixing

## 3.1 Langevin Dynamics

**Figure 3.3: Illustration of Langevin sampling.** Langevin sampling using the score function  $\nabla_{\mathbf{x}} \log p_{\phi}(\mathbf{x})$  to guide trajectories toward high-density regions via the update in Equation (3.1.5) (indicating by arrows).

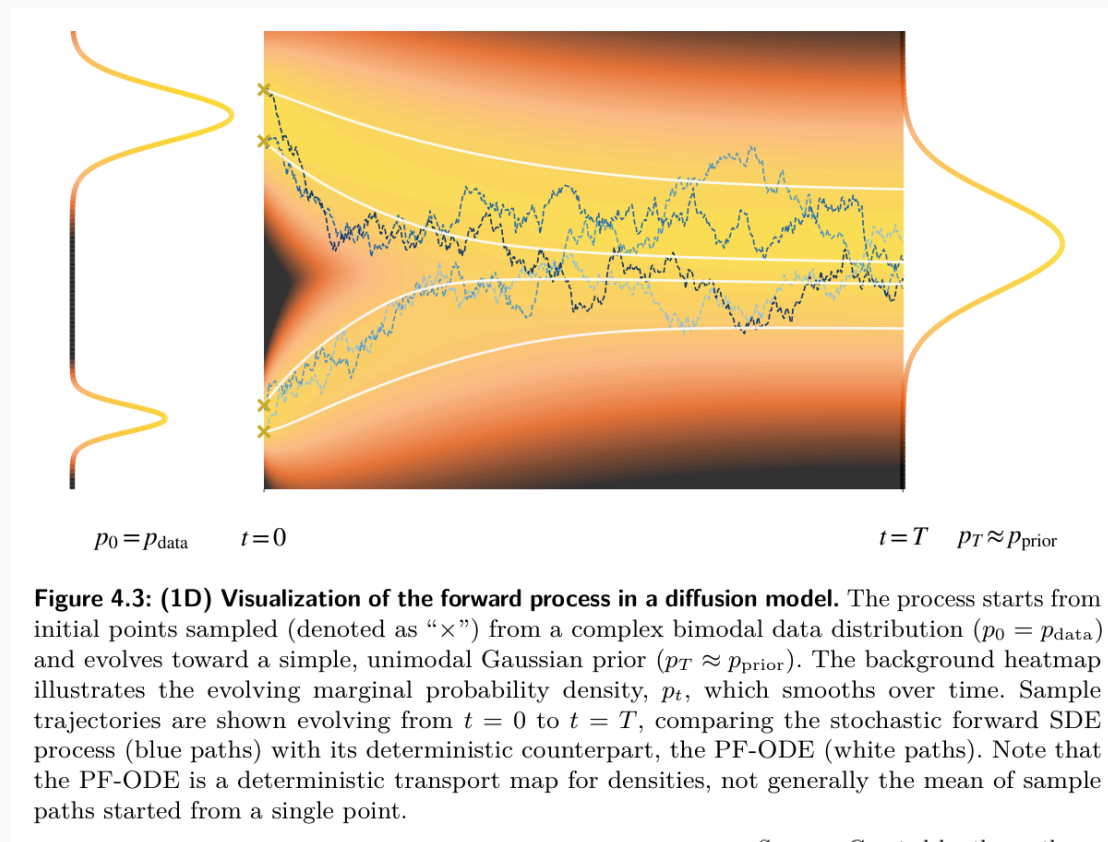
## 4 · The Continuous-Time View



## 4.1 The Forward SDE

- Let steps  $\rightarrow 0$ : the noising chain becomes an SDE
- $dx = f(x, t) dt + g(t) dw$  (drift + diffusion)
- VP-SDE corresponds to DDPM; VE-SDE to NCSN
- Marginal stays  $p_t(x_t|x_0) = N(\alpha_t x_0, \sigma_t^2 I)$
- A whole **family** of distributions  $p_t$ , indexed by time

## 4.1 The Forward SDE

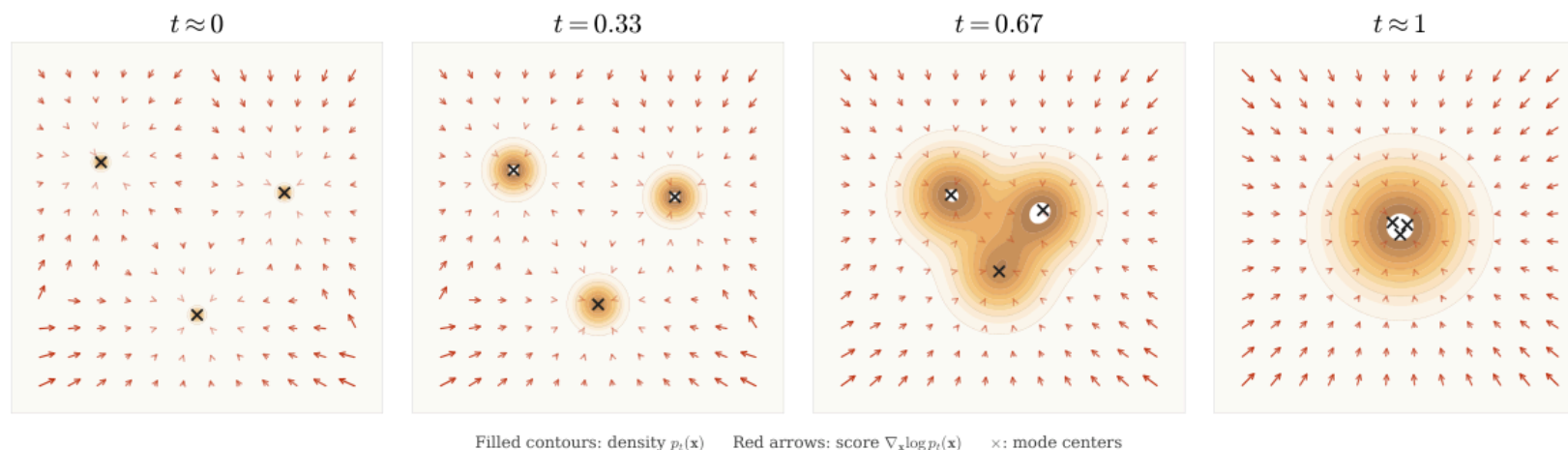


## 4.2 The Time-Dependent Score

- Now the score depends on time:  $s(x, t) = \nabla_x \log p_t(x)$
- One network  $s_\theta(x, t)$  for all noise levels
- Smoothed at high  $t$  (easy), sharp at low  $t$  (hard)
- This is exactly the NCSN idea, in continuous time
- Trained by denoising score matching at each  $t$



## 4.2 The Time-Dependent Score

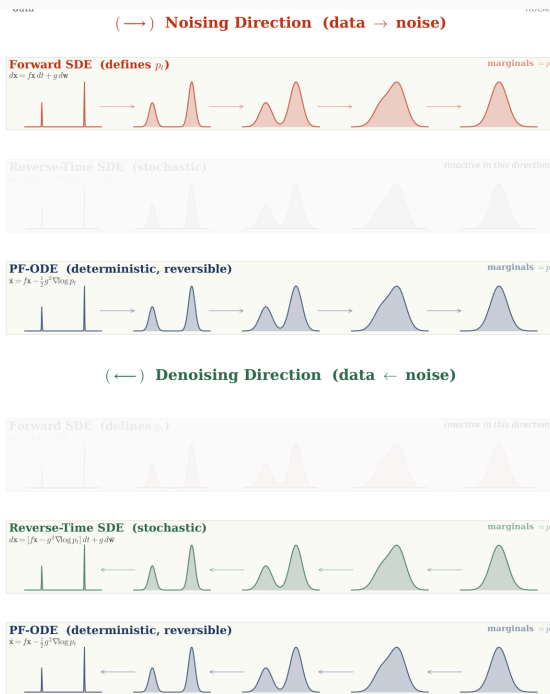


**Figure 4.1: Illustration of the time-dependent score landscape.** As the noise level increases, the perturbed density  $p_t(\mathbf{x})$  evolves from a multimodal distribution (data) concentrated near the data modes to a broad nearly unimodal distribution (prior). Accordingly, the score field  $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$  changes from a sharply local vector field that pulls samples toward nearby modes to a smooth global field that points toward the overall center of mass. This time-dependent score field is the key object learned in score-based diffusion models (i.e., Score SDE).

## 4.3 Heuristic: Reversing an SDE

- Chain rule:  $p_{t,\delta}(x, y) = p_{t+\delta|t}(y|x)p_{t(x)} = p_{t|t+\delta}(x|y)p_{t+\delta}(y)$
- Forward step (Euler):  $p_{t+\delta|t}(y|x) = N(y \mid x + f^+\delta, \delta\sigma^2)$
- Rearrange:  $p_{t|t+\delta}(x|y) = p_{t+\delta|t}(y|x) \frac{p_{t(x)}}{p_{t+\delta}}(y)$
- Taylor on  $\log p_t$  around  $y$ : factor  $\exp\left((x - y)^T \nabla_y \log p_{t(y)}\right)$
- Complete the square  $\rightarrow$  reverse drift  $f^- = -f^+ + \sigma^2 \nabla \log p$
- Score appears because reversing needs the log-density gradient

## 4.3 Heuristic: Reversing an SDE



**Figure 4.5: Three dynamics sharing the same marginals  $p_t$ .** The forward SDE, which serves as an anchor reference, defines the marginals  $p_t$  by diffusing data toward a Gaussian prior. The reverse-time SDE and the PF-ODE yield the same marginals, as guaranteed by the Fokker-Planck equation, while moving in the denoising direction through the score  $\nabla \log p_t$ . The top panel shows noising, given by the forward SDE or, equivalently at the marginal level, the PF-ODE run forward in time. The bottom panel shows denoising, given by the reverse-time SDE or the PF-ODE run backward in time. The PF-ODE appears in both panels because it is deterministic and time-reversible.

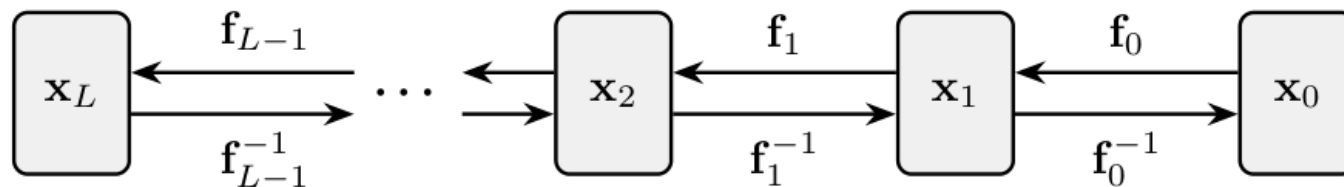
## 4.4 Reverse SDE & Probability-Flow ODE

- Anderson (1982): rigorous reverse-time SDE with score term
- $dx = [f - g^2 \nabla \log p_t] dt + g d\bar{w}$
- Probability-flow ODE: **same marginals**, no noise
- $dx = [f - \frac{1}{2}g^2 \nabla \log p_t] dt$
- SDE = stochastic sampler; ODE = deterministic + likelihood
- Full PF-ODE derivation from FPE: Day 8

## 4.5 Derivation: The DSM Objective

- Want  $s_\theta(x, t) \approx \nabla \log p_t(x)$  (marginal)
- Marginal score =  $\mathbb{E}[\nabla \log p_t(x_t \mid x_0) \mid x_t]$
- So regress on the **conditional** score (known Gaussian)
- $\nabla \log p_t(x_t \mid x_0) = -(x_t - \alpha_t x_0) \frac{1}{\sigma_t^2} = -\varepsilon \frac{1}{\sigma_t}$
- $\rightarrow$  predicting the score = predicting the noise

## 4.5 Derivation: The DSM Objective



**Figure 5.2: Illustration of a NF.** An NF consists of a stack of invertible maps  $\mathbf{f}_\phi = \mathbf{f}_{L-1} \circ \mathbf{f}_{L-2} \circ \dots \circ \mathbf{f}_0$ . The transformation maps latent samples  $\mathbf{x}_0 \sim p_{\text{prior}}$  to data samples  $\mathbf{x}_L \sim p_{\text{data}}$ .

*Source: Created by the authors*

## 5 · Flow Matching

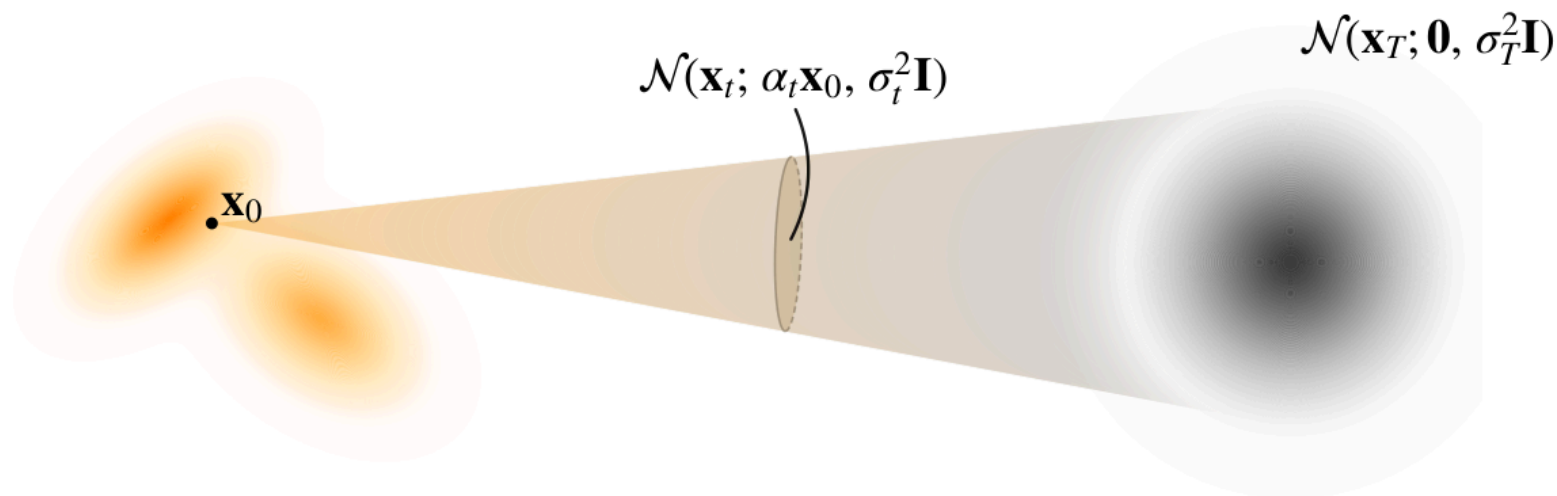
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## 5.1 Continuous Normalizing Flows

- Transport noise to data along an ODE:  $dx = v_\theta(x, t) dt$
- $v$  = a **velocity field** moving samples in time
- Density evolves by the continuity equation
- Old way (CNF): expensive max-likelihood training
- Flow matching: regress  $v$  directly, simulation-free



## 5.1 Continuous Normalizing Flows



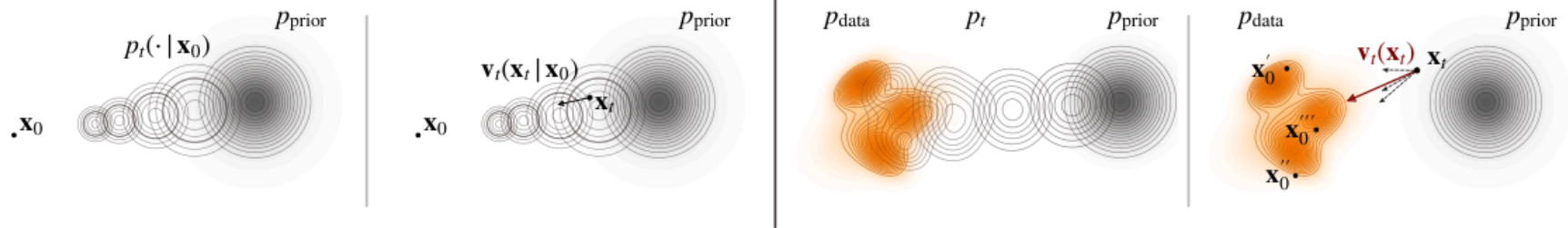
**Figure 5.5: Illustration of the conditional transition distribution.**  $p_t(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \alpha_t \mathbf{x}_0, \sigma_t^2 \mathbf{I})$ , defines a (Gaussian) conditional probability path from a data sample  $\mathbf{x}_0 \sim p_{\text{data}}$  (left) towards the Gaussian prior  $p_{\text{prior}}$  (right).

*Source: Created by the authors*

## 5.2 Conditional Flow Matching

- Pick a simple per-sample path, e.g.  $x_t = (1 - t)x_0 + tx_1$
- Its velocity is known:  $u_t(x \mid x_1) = x_1 - x_0$
- Regress  $v_\theta(x_t, t)$  onto this conditional velocity
- $J_{\text{CFM}} = \mathbb{E} \lVert v_\theta(x_t, t) - u_t(x \mid x_1) \rVert^2$
- No simulation, no divergence term — just regression

## 5.2 Conditional Flow Matching

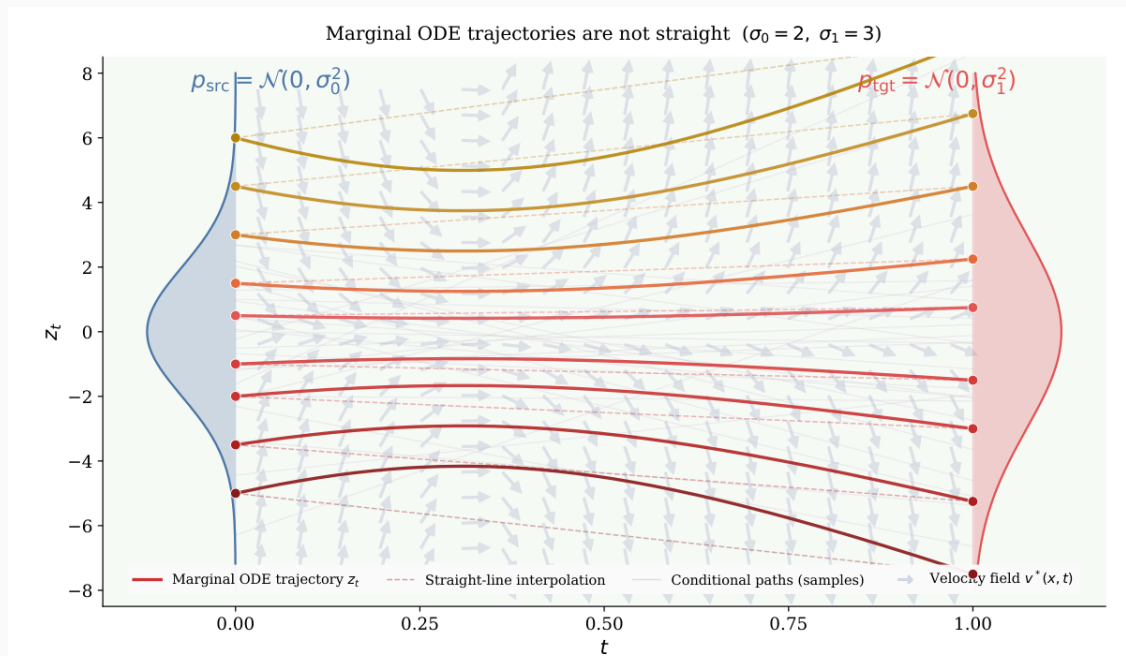


**Figure 5.6: Illustration of conditional versus marginal perspectives in diffusion.** (This figure is motivated by Lipman *et al.* (2024).) (1) Conditional Gaussian path  $p_t(\cdot|\mathbf{x}_0)$ , showing expanding densities from a fixed  $\mathbf{x}_0$  toward the prior. (2) Conditional velocity  $\mathbf{v}_t(\mathbf{x}_t|\mathbf{x}_0) = f(t)\mathbf{x}_t - \frac{1}{2}g^2(t)\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t|\mathbf{x}_0)$ . (3) Marginal density  $p_t$ , transporting the data distribution  $p_{\text{data}}$  (orange) into the prior  $p_{\text{prior}}$  (gray). (4) Marginal velocity  $\mathbf{v}_t(\mathbf{x}_t) = f(t)\mathbf{x}_t - \frac{1}{2}g^2(t)\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$ .

## 5.3 Conditional vs Marginal Velocity

- Many conditional paths overlap at a point  $x_t$
- Marginal velocity = average of conditionals through  $x_t$
- $v(x, t) = \mathbb{E}[u_t(x \mid x_1) \mid x_t = x]$
- CFM and marginal FM share the **same** gradient
- So regressing conditionals learns the marginal field

## 5.3 Conditional vs Marginal Velocity

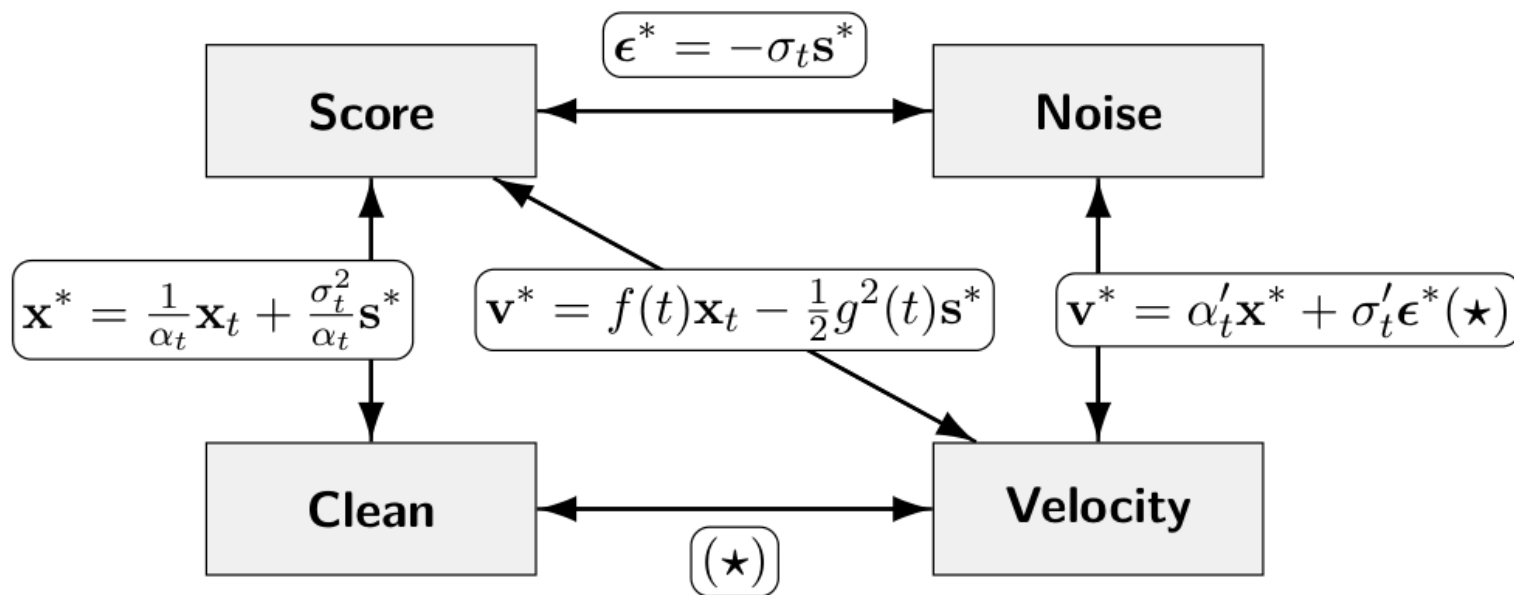


**Figure 5.9: Marginal ODE trajectories are not straight under the canonical affine interpolation.** We take  $x_0 \sim \mathcal{N}(0, \sigma_0^2)$  and  $x_1 \sim \mathcal{N}(0, \sigma_1^2)$  with  $\sigma_0 = 2, \sigma_1 = 3$ , and independent coupling. Gray lines show sampled conditional paths  $(1-t)x_0 + tx_1$ , which are straight by construction. Solid curves show the exact marginal ODE trajectories  $z_t = z_0 \sqrt{(1-t)^2 + (\sigma_1/\sigma_0)^2 t^2}$ , and dashed lines show the corresponding straight-line interpolations from  $z_0$  to  $z_1$ . The visible gap between

## 5.4 Rectified Flow & Reflow

- Linear interpolation  $\rightarrow$  straight conditional paths
- But **marginal** trajectories can still be curved
- Curved paths need many ODE steps to integrate
- Reflow: retrain on (noise, sample) pairs  $\rightarrow$  straighter
- Straighter flow  $\rightarrow$  fewer steps, even one-step

## 5.4 Rectified Flow & Reflow



**Figure 6.1: Equivalent relations among four parameterizations.**  $\mathbf{v}$ -prediction is given by  $\mathbf{v}^* = \alpha'_t \mathbf{x}^* + \sigma'_t \epsilon^*$ , where clean and  $\epsilon$ -predictions are interchangeable via  $\mathbf{x}_t = \alpha_t \mathbf{x}^* + \sigma_t \epsilon^*$ .

Source: Created by the authors

## 6 · One Model, Many Views

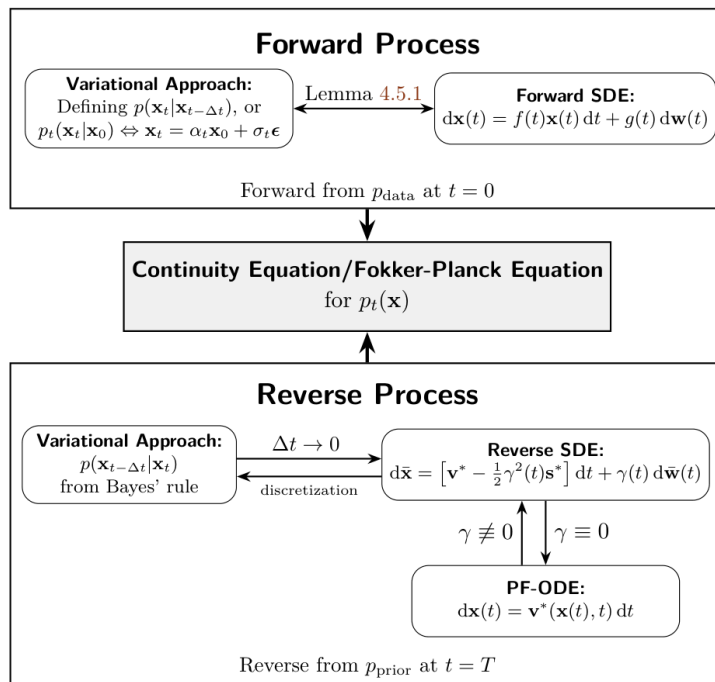
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## 6.1 Four Equivalent Parameterizations

- Predict noise  $\varepsilon$ , data  $x_0$ , score  $s$ , or velocity  $v$
- All related by the forward rule  $x_t = \alpha_t x_0 + \sigma_t \varepsilon$
- $s = -\varepsilon \frac{1}{\sigma_t}$  (score = scaled noise)
- $v = \alpha'_t x_0 + \sigma'_t \varepsilon$  (flow-matching velocity)
- Same network, different target — pick for stability

# 6.1 Four Equivalent Parameterizations



**Figure 6.2: Unified perspective connecting variational, SDE, and ODE formulations through the continuity equation, where all  $p_t(\mathbf{x})$  evolve under a shared dynamic.** The velocity field  $\mathbf{v}^*(\mathbf{x}, t) = f(t)\mathbf{x} - \frac{1}{2}g^2(t)\mathbf{s}^*(\mathbf{x}, t)$  is governed by the score function  $\mathbf{s}^*(\mathbf{x}, t) := \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ . Coefficients  $f(t)$ ,  $g(t)$ ,  $\sigma_t$ , and  $\alpha_t$  are pre-defined time-dependent functions, and  $\gamma(t)$  is a tunable

## 6.2 The Unified Picture

- Variational (DDPM), score-SDE, and flow matching coincide
- All learn the same time-indexed field over  $p_t$
- Sample with: ancestral, Langevin, reverse SDE, or ODE
- Score  $\leftrightarrow$  noise  $\leftrightarrow$  velocity are interchangeable
- Day 8: guidance, fast solvers, and few-step sampling

- Day 7: **Score, SDEs & Flow Matching**
- The continuous-time view of diffusion
- Questions welcome — practical follows

## Questions?

Practical session follows — see course site.